

SWETHALATHA SIVAKUMAR

Software & IoT Engineering Aspirant

Coimbatore, India

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Professional Summary

Motivated Computer Science and Engineering (IoT) undergraduate with strong foundations in programming, IoT systems, and modern web development. Experienced in building real-time GPS-integrated automation solutions, AI-based applications, and embedded systems. IEEE international conference author, patent holder, and national-level hackathon awardee with proven problem-solving, leadership, and teamwork abilities.

Education

Bachelor of Engineering – Computer Science & Engineering (IoT)

Nandha Engineering College

2023 – 2027

CGPA: **8.9**

Higher Secondary Education

K.V Higher Secondary & CBSE School

Percentage: **82%**

Technical Skills

- **Programming Languages:** C, C++, Python, Java
- **Web Technologies:** HTML, CSS, JavaScript
- **IoT & Embedded Systems:** Sensors, Arduino, Raspberry Pi, GPS Integration
- **Tools & Platforms:** Git, VS Code, Canva
- **Core Competencies:** Problem Solving, Analytical Thinking, Team Leadership, Effective Communication

Projects

Smart Retractable Speed Breaker System with GPS Module

- Designed an automated traffic management system using GPS-based ambulance detection.
- Enabled dynamic speed breaker control for smooth emergency vehicle passage.
- Improved road safety and traffic efficiency through real-time system response.

Weather Prediction Website

- Developed a responsive web application providing real-time weather updates using API integration.
- Designed a user-friendly interface optimized for multiple devices.

AI Personalized Chatbot

- Built an AI-powered chatbot capable of understanding user queries and delivering context-aware responses.
- Integrated frontend UI with intelligent backend logic for smooth conversational experience.

- Focused on user personalization, response accuracy, and modern UI/UX design.

Alcohol Detection and Engine Locking System with GPS Tracking

- Designed an embedded safety system to detect alcohol consumption and automatically lock the vehicle engine.
- Integrated GPS tracking for real-time vehicle location monitoring.
- Implemented using sensors, microcontrollers, and tracking modules.

Research & Publications

Smart Retractable Speed Breaker System with GPS Module

- Accepted and presented at the **6th International Conference on Electronics and Sustainable Communication Systems (ICESCS-2025)**.
- Conducted at Hindusthan Institute of Technology, Coimbatore (Sept 10–12, 2025).
- Indexed in IEEE Xplore.

Design and Implementation of an Intelligent Face Attendance Recognition System Using Deep Learning and Computer Vision Techniques

- Presented at an international conference.
- Implemented deep learning and computer vision techniques for automated attendance systems.

Patent

EMG Silent Speech and Biometric Control System for Smart IoT Devices Powered by Body Heat

- Proposed an EMG-based silent speech control system combined with biometric authentication.
- Utilized body heat as an alternative power source for sustainable IoT operation.
- Designed for secure, hands-free, and eco-friendly human–device interaction.

Internship Experience

Inaiyam Innovations — Tech Associate Intern (Hybrid) Oct 6 – Jan 6

- Completed a 3-month hybrid internship focusing on technical and system-level problem solving.
- Gained exposure to real-world development workflows and collaborative environments.

AmyPo Technologies — Intern 2024 – 2025

- Worked on real-time IoT and software development projects.
- Enhanced practical debugging, implementation, and teamwork skills.

Achievements

- **Best All-Girls Team Award – PEC HACKS 3.0 (36-Hour Hackathon)**
- Awarded **35,000 cash prize** for innovative technical solution.
- MIT Chennai – Greek Language (Coding) Competition: **2nd Place**

Languages

Tamil, English, Hindi